Model answer to Q 1(a): Applied Bookwork and Discussions in Class

This question relates to power link budgets.

i) A single-stage subsystem has a power gain G and noise factor F. Using the well-known relationship for the noise factor, in terms of signal-to-noise ratios (assume room-temperature conditions), prove that the output noise power $No = G \cdot F \cdot Ni$.

$$F = \frac{\left({}^{Si}/_{Ni}\right)}{\left({}^{So}/_{No}\right)} = \frac{No}{GNi} \text{ where } G = {}^{So}/_{Si} \text{ and so } No = G \cdot F \cdot Ni$$

[2]

ii) For a subsystem with three cascaded stages, give the expression for output noise power in terms of the individual noise power contributions and the individual power gains for the stages. Using this expression, derive the corresponding Friis' Formula for the three-stage subsystem, in terms of the individual noise factors and their power gain.

$$No = (Ni + N_1)G_1G_2G_3 + N_2G_2G_3 + N_3G_3 \text{ for 3 stages in cascade}$$

$$F = \frac{(Ni+N_1)G_1G_2G_3+N_2G_2G_3+N_3G_3}{GN_i} \text{ where } G = G_1G_2G_3$$

$$N_i = kToB;\ N_1 = kTo(F_1 - 1)B;\ N_2 = kTo(F_2 - 1)B;\ and\ N_3 = kTo(F_3 - 1)B$$

$$\therefore F = F_1 + \frac{(F_2-1)}{G_1} + \frac{(F_3-1)}{G_1G_2} \text{ Friis' Formula}$$

[6]

iii) If the first stage of multi-stage cascaded subsystem employs an LNA, using the Friis' Formula, justify why this first stage dominates the complete subsystem in terms of noise performance.

An LNA has a low noise figure, but equally important is that it has a high power gain. As a result, with a large G_1, the second term is not significant and any following terms are negligible. Therefore, with the LNA as the first stage, it dominates the noise contribution introduced by the subsystem.

[2]

iv) Why are LNAs usually much more expensive than the following stages in a receiver subsystem? Briefly explain why LNAs are not generally found in silicon integrated circuits.

Modern LNA are made from exotic semiconductors (e.g. InP/pHEMT fabricated on an Al-GaAs wafer). This technology requires a great deal more processing that standard GaAs MESFET or Si/SiGe wafer processing. For this reason, silicon integrated circuit fabrication processing is not compatible with InP/pHEMT processing and so the LNA is usually found as a discrete device at the input of a silicon RFIC.

[2]

v) What is the generally accepted room temperature (in Kelvin) used within standard test laboratories? Comment on how this compares to temperatures that have to meet military standards and/or in space environments.

Answers

23°C or 296 K is a standard ambient room temperature found in metrology laboratories. In contrast, −55℃ *to* +125℃ for Military specifications; and space can go down to −270℃.

[2]

Model answer to Q 1(b): Applied Bookwork and Discussions in Class

Consider the LNB shown in Figure 1.

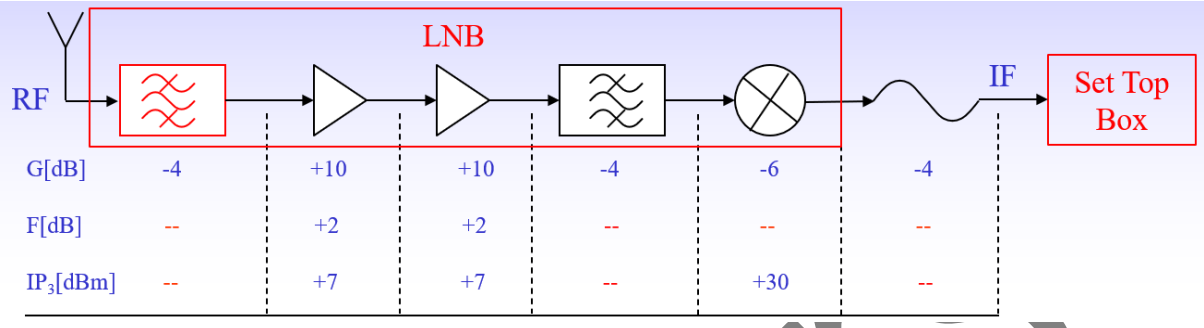


Figure 1.

i) Spell out the abbreviation LNB.

[2]

ii) What domestic application uses an LNB and state the approximate frequency ranges for the RF input and IF output.

Satellite TV is often used around the world, especially in remote locations. The downlink RF is approximately between 11.70 and 12.75 GHz (X-band) and, with a 10.70 GHz LO, the corresponding IF frequency range is 1,000 to 2,050 MHz.

[2]

iii) What is the normal role of the first BPF in the LNB chain (in red).

The first bandpass filter is often used to filter-out any out-of-band interfering signals and noise.

[2]

iv) What is the normal role of the second BPF in the LNB chain.

The second bandpass filter is used as an image-rejection filter for the following mixer. This is usually required as the first BPF may have a low order and so let in some of the image frequency spectrum – which would otherwise cause interference at the baseband.

[2]

v) Given that the mixer has a conversion loss, what does this suggest about its implementation (i.e., is it active or passive?). State your reasoning.

Given a conversion loss of 6 dB, this suggests that the mixer is passive (i.e., has no active transistors). Moreover, with such a high loss, the mixer will have a balanced or even double-balanced topology.

[2]

vi) What is a suitable IP3 value for the passive components? State your reasoning.

In practice, passive components are always linear (unless operated at very high power levels), but an infinite value of IP3 is not very useful for doing power link budget calculations. For this reason, a value of +100 dBm is a useful IP3 value (albeit not the actual value) found in spreadsheet calculations.

[2]

Answers

Model answer to Q 1(c): New Calculations

From Figure 1, showing all your working out, calculate the following at the output of each stage:

i) Gain (in dB).

$G = G_1 G_2 G_3$ for a 3-stage example

G[dB]	-4	+6	+16	+12	+6	+2

[4]

ii) Noise Figure (in dB).

$F = F_1 + \frac{(F_2-1)}{G_1} + \frac{(F_3-1)}{G_1 G_2}$ for a 3-stage example

F[dB]	+4.0	+6.0	+6.2	+6.2	+6.4	+6.8

[8]

iii) Third-order Intercept Point (in dBm).

$$IP3_{RX} = (IP3_1 \cdot G_2 \cdot G_3) // (IP3_2 G_3) // (IP3_3)$$

$$\therefore IP3_{RX} = \frac{1}{\frac{1}{IP3_1 \cdot G_2 \cdot G_3} + \frac{1}{IP3_2 G_3} + \frac{1}{IP3_3}}$$

for a 3-stage example

IP_3[dBm]	+100	+7.0	+6.6	+2.6	+3.4	-7.4

[8]

iv) If the first BPF (in red) is completely removed, comment on what you would expect to see in the overall Gain, Noise Figure and Third-order Intercept Point.

By removing the first LNB filter, the overall gain increases to 4.0 dB, the noise figure will significantly reduce (only be 2.75 dB) and the IP_3 stays the same (at 7.4 dBm).

[4]

Model answer to Q 2(a): Textbook for New Application

Consider the ideal 5-port network illustrated in Figure 2, consisting of only three couplers and four unspecified square blocks.

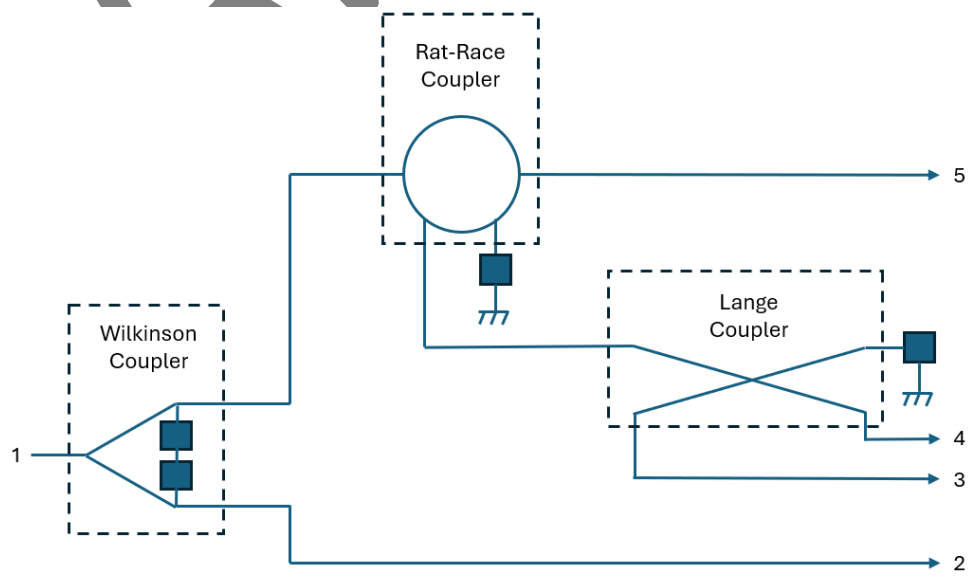
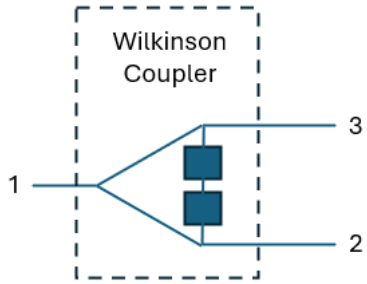


Figure 2.

Answers

Sketch the individual couplers and clearly indicate each port number. Write down the ideal S-parameter matrices for each of the ideal couplers operating at their centre frequency:

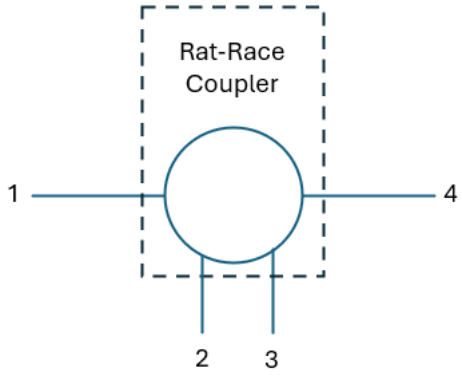
(i) Wilkinson.



$$[S]_{Wilkinson} = \frac{-j}{\sqrt{2}}\begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix} @\ \omega_0$$

[2]

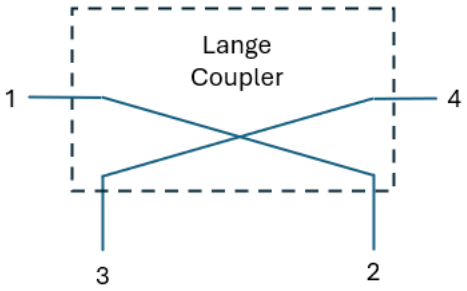
(ii) Rat-Race.



$$[S]_{Rat-Race} = \frac{-j}{\sqrt{2}}\begin{pmatrix} 0 & 1 & 0 & -1 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ -1 & 0 & 1 & 0 \end{pmatrix} @\ \omega_0$$

[2]

(iii) Lange.



$$[S]_{Lange} = \frac{-j}{\sqrt{2}}\begin{pmatrix} 0 & 1 & j & 0 \\ 1 & 0 & 0 & j \\ j & 0 & 0 & 1 \\ 0 & j & 1 & 0 \end{pmatrix} @\ \omega_0$$

[2]

Answers

Model answer to Q 2(b): Textbook for New Application

What do the square (filled-in) boxes represent? With those associated with the Rat-Race and Lange couplers, what are they usually called and what is the ideal S-parameter matrix for these port terminations?

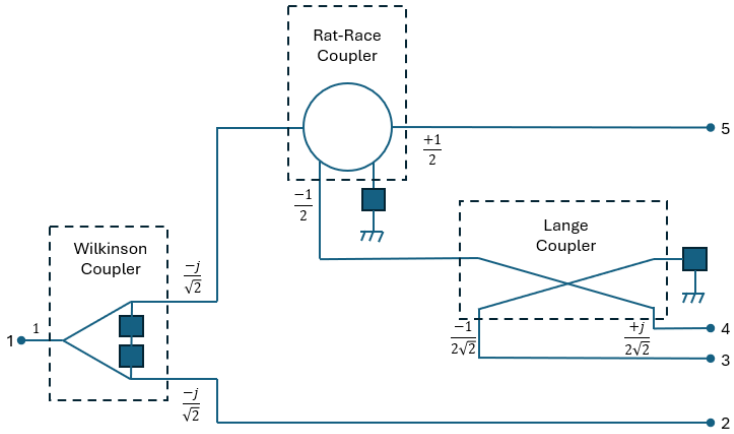
The boxes represent resistors of value Zo. With the Rat-Race and Lange couplers, these resistors are called "dummy loads" and they have an ideal S-parameter matrix of $[S] = (0)$.

[3]

Model answer to Q 2(c): Derivation for New Application

By inspection, determine the ideal 5×5 S-parameter matrix for the complete 5-port network. Hint: start by clearly state all the simplifying assumptions.

Assume all resistors have a value of *Zo*. In addition, all couplers are lossless, perfectly impedance matched to *Zo* and that they have infinite isolation characteristics. Finally, all coupler interconnects are lossless, perfectly impedance matched and have zero transmission phase angles.



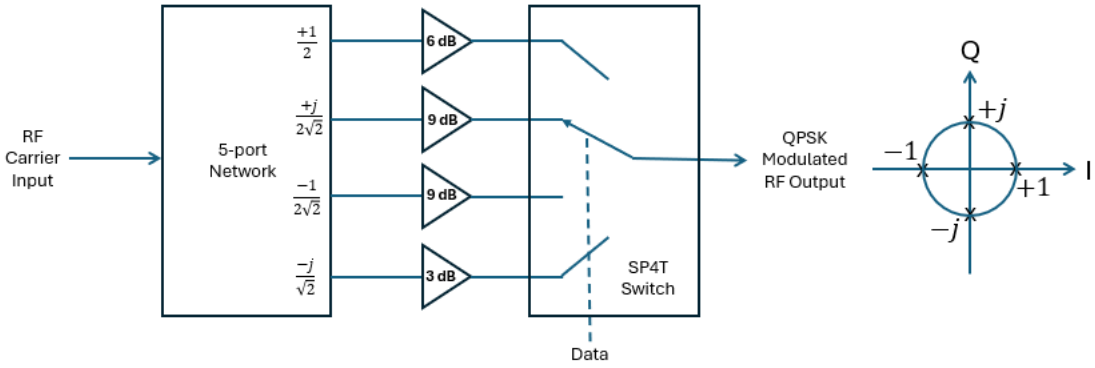
$$[S] = \frac{-j}{\sqrt{2}} \begin{pmatrix} 0 & 1 & \frac{-j}{2} & \frac{-1}{2} & \frac{+j}{\sqrt{2}} \\ 1 & 0 & 0 & 0 & 0 \\ \frac{-j}{2} & 0 & 0 & 0 & 0 \\ \frac{-1}{2} & 0 & 0 & 0 & 0 \\ \frac{+j}{\sqrt{2}} & 0 & 0 & 0 & 0 \end{pmatrix}$$

[9]

Model answer to Q 2(d): New Design

Given the 5-port network shown in Figure 2, four amplifiers and a data-driven single-pole four-throw (SP4T) switch, design a simple QPSK modulator and draw the resulting I-Q constellation. The gain of the individual amplifiers should be chosen so that all the QPSK symbols are located on a unit circle, centred on the origin of the I-Q chart. Hint: start by clearly state all the simplifying assumptions.

Assume all amplifiers, the SP5T switch and all interconnects are lossless, perfectly impedance matched and have zero insertion phase angles.



[7]

ANSWERS

Model answer to Q 3(a): New Derivation

Derive, from first principles, the general radar range equation and include all units. Assume the following lossless system:

- The transmitter has an output power P_T feeding an antenna having a power gain G_T that is located at a range R_T from the target.
- The target has a radar cross section σ.
- The receiver has an input power P_R delivered by an antenna having a power gain G_R that is located at a range R_R from the target.

At the target, power density is given by:

$$PD_{TARGET} = \left(\frac{P_T}{4\pi R_T^2}\right) G_T \quad [\mathrm{W/m^2}]$$

Power captured and re-radiated isotropically by the target, $P_{TARGET} = \sigma PD_{TARGET}$ [W]

At the receiver, power density is given by:

$$PD_{RX} = \left(\frac{P_{TARGET}}{4\pi R_T^2}\right) \quad [\mathrm{W/m^2}]$$

Power at the input to the receiver:

$$P_R = A_{RX} PD_{RX} \quad [\mathrm{W}]$$

The effective aperture of the receiving antenna, $A_{RX} = \frac{\lambda_o^2}{4\pi} D_o$ $[\mathrm{m^2}]$

Where $G_R = D_o \eta$ and with a lossless antenna its efficiency, η = 100%

$$\therefore P_R = \left(\frac{\lambda_o^2}{4\pi}\right) G_R \left(\frac{1}{4\pi R_R^2}\right) \sigma \left(\frac{P_T}{4\pi R_T^2}\right) G_T$$

$$\therefore \left(\frac{P_R}{P_T}\right) = \left(\frac{\lambda_o}{4\pi R_T R_R}\right)^2 \frac{G_T \sigma G_R}{4\pi} \quad \text{Radar Range Equation}$$

When the transmitter and receiver share a common antenna:

$$\left(\frac{P_R}{P_T}\right) \rightarrow \left(\frac{\lambda_o G}{4\pi R^2}\right)^2 \frac{\sigma}{4\pi}$$

[5]

Model answer to Q 3(b): Calculated Example

Two amateur radio enthusiasts, living 1 km apart, communicate at 432 MHz using the overhead Moon as a passive satellite (i.e., a reflecting target). The distance to the Moon from both enthusiasts is R_M =

Answers

381,500 km and the diameter of the Moon is D_M = 3,500 km. Neglect the effects of the Earth's atmosphere.

i) If identical lossless paraboloidal reflector antennas are used at both locations, calculate the power gain in dBi of the antennas if they have a diameter of 9 m. Note that for an ideal lossless paraboloidal reflector antenna its directivity D_o is given by:

$$D_o = 4\pi \frac{Aperture\ Area}{\lambda^2} \tag{3.1}$$

$$A = \frac{\pi D^2}{4} = 63.62 \quad \text{m}^2$$

$$\Omega = \frac{\lambda_o^2}{A} = 7.6x10^{-3} \quad with \quad \lambda_o = 0.694 \text{ m}$$

$$Directivity,\ D_o = \frac{4\pi}{\Omega} = 1{,}660$$

$$Power\ Gain,\ G \rightarrow D_o = 32.2 \text{ dBi}$$

[3]

ii) Comment on the resulting beam efficiency for this application, when considering the angle that the Moon subtends as seen by an observer on the Earth, if the -3 dB beam width θ is given by:

$$\theta = \sqrt{\frac{4\pi}{D_o}} \tag{3.2}$$

With a pencil beam radiation pattern (i.e. having a large aperture and low sidelobes):

$$\Omega \approx \theta_E \theta_H,$$

Where θ_E and θ_H are the -3 dB beamwidths in the E- and H-planes, respectively.

$$\therefore \theta \approx \sqrt{\Omega} = 0.087\ radians \equiv 5^o$$

To an observer on the Earth, the Moon subtends an angle of 2ϕ: where

$$2\phi = 2\ tan^{-1}\left(\frac{3.5/2}{381.5 + 3.5/2}\right) = 0.523^o$$

Therefore, the beam efficiency will be very small because the angle subtended by the target is an order of magnitude smaller than the beamwidth of the antenna.

[3]

iii) With P_T = 20 dBW, calculate the power at the receiver. As a first-order approximation, assume that the Moon's radar cross section can be modelled as a perfectly reflecting flat circular disc. Neglect ohmic and diffraction losses.

If we assume that the Moon's radar cross-section can be modelled as a perfectly reflecting flat circular disc:

$$\sigma = \frac{\pi D_M^2}{4} = 9.62x10^{12} \quad \text{m}^2$$

Answers

Therefore, the power at the receiver is given by:

$$P_R = \left(\frac{\lambda_o D_o}{4\pi R^2}\right)^2 \frac{\sigma P_T}{4\pi} = 30 \text{ aW}$$

[3]

iv) With an antenna temperature T_A = 100 K and a receiver noise temperature T_{RX} = 75 K, calculate the system noise temperature T_S, input $(G_R\ /T_S)$ and output (S/N) at a receiver having a final IF bandwidth of 7 kHz.
Boltzmann's constant k = 1.38 x 10^{-23} W/Hz/K.

System equivalent noise temperature at the receiver, $T_S = T_A + T_{RX} = 175$ K

$$\left(\frac{G_R}{T_S}\right) = \frac{1{,}660}{175} \rightarrow 9.6 \text{ dB/K}$$

System Output Equivalent Noise Power, $N = kT_S\ B = 16.9$ aW

Therefore, the output signal-to-noise power ratio at a receiver, $S/N = P_R/N = 1.775 = 2.5$ dB

[8]

v) Calculate the minimum theoretically possible propagation delay time between the two ground stations if both antennas suffer from unwanted side lobes.

If both antennas suffer from unwanted sidelobes then it is theoretically possible that the sensitive receiver could detect the high power transmitter through unwanted mutual coupling of the sidelobes from both antennas. In this case, the minimum possible delay will be given from:

Delay = separation distance between Earth stations / speed of light = 1,000/300,000,000 = 3.33 μs.

[3]

Model answer to Q 4(a): New Application of Theory

A lumped-element *L-C* band stop filter (BSF) has to meet the following specifications:

Lower-pass band -3 dB cut-off frequency:	9.25 GHz
Upper-pass band -3 dB cut-off frequency:	10.75 GHz
Worst-case pass-band insertion loss:	0.1 dB
Band-stop attenuation:	> 110 dB
Stop-band bandwidth:	150 MHz
Source impedance, *Zs*:	25 Ω
Load impedance, Z_L:	50 Ω

Answers

Calculate the centre frequency, pass-band bandwidth *Bp* and prototype *f/fc* ratio.

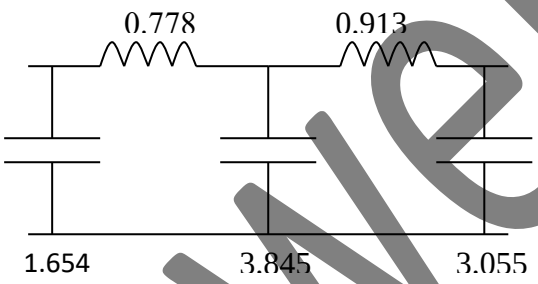
-3 dB cut-off frequencies are at $f_{P1} = 9.25$ GHz and $f_{P2} = 10.75$ GHz
Centre frequency, $fo = \sqrt{(f_{P1} f_{P2})} = 9.97$ GHz
Pass band bandwidth, $Bp = f_{P1} - f_{P2} = 1.5$ GHz
Stop band bandwidth, $Bs = 150$ MHz
$f/fc = Bp/Bs = 10$

[5]

Model answer to Q 4(d): New Application of Theory

From the standard filter curves and tables supplied, determine the normalised component values for the low-pass prototype that meet the required specification. Justify your choice of curve and table.

From the attenuation curves, the 5th order 0.1 dB ripple Chebyshev filter with $Rs/R_L = 0.5$ meets the specification with a stop band attenuation of 113 dB (i.e. a margin of 3 dB). The normalised values for the low-pass prototype is given below:



[5]

Model answer to Q 4(c): New Application of Theory

Draw the topology for the BPF and calculate the component values.

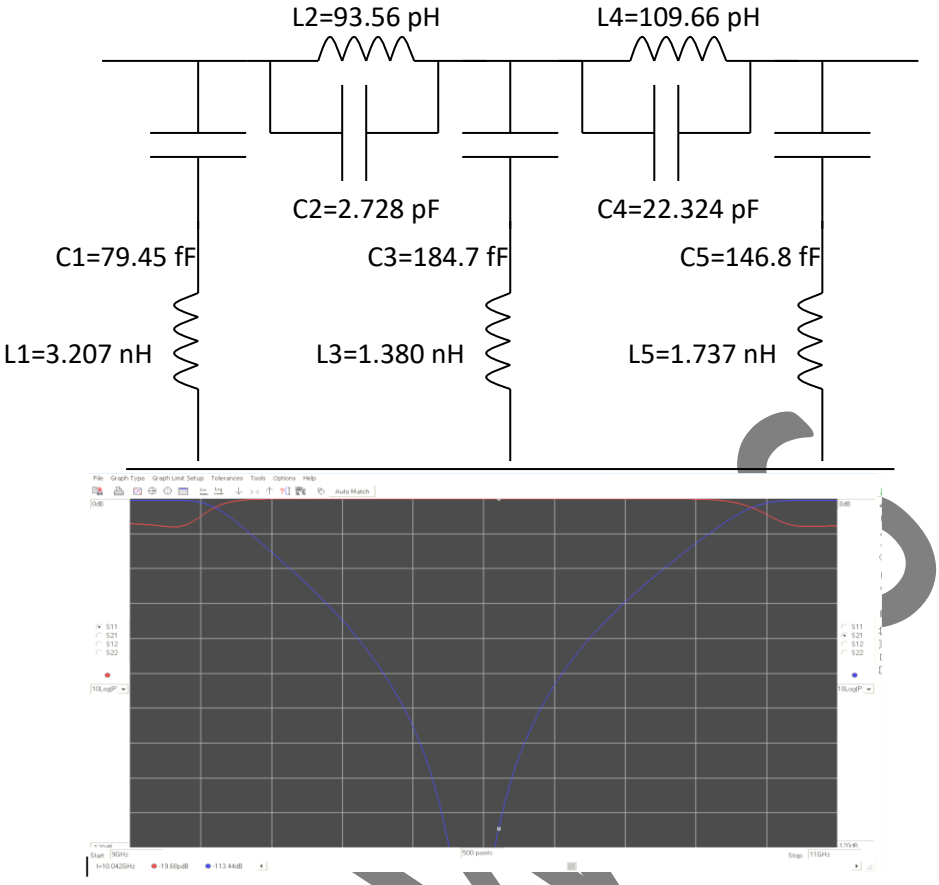
For a band stop filter, the capacitors are replaced by a series *L-C* resonators and the inductors are replaced by parallel *L-C* resonators.
The un-normalized shunt connected series tuned circuit element values are:

$$Cs = \frac{BpCn}{2\pi f_o^2 R_L} \quad and \quad Ls = \frac{R_L}{2\pi Bp L_n}$$

The un-normalized series connected parallel tuned circuit element values are:

$$Cp = \frac{1}{2\pi BpCnR_L} \quad and \quad Lp = \frac{BpLnR_L}{2\pi f_o^2}$$

Answers



[10]

Model answer to Q 4(d): New Application of Theory

Comment on the range of component values and suggest more practical methods for implementing the filter.

The inductance values range from 94 pH to 3.2 nH, while the capacitors range from 79 fF to 2.7 pF, a ratio of 34:1 in both cases. This makes it very difficult to achieve such a large range with the same technology. Moreover, the precision to 3 decimal places is also impractical.

To solve the first problem, impedance/admittance inverters can be employed. While distributed-element components can be employed to replace the more sensitive lumped-element components.

[5]

Answers